🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 · STANDARD 6.NOS.4

Prime Factorization

Lesson 1-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes **Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write a number as a product of its prime factors using a factor tree.

Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Prime number | **Composite number** | **Prime factorization** | **Factor** | **Factor tree** | **Exponent**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A whole number greater than 1 with exactly two factors, 1 and itself, is a ________ number.

2. A whole number greater than 1 that has more than two factors is a ________ number.

3. ________ — Writing a number as prime numbers multiplied together.

4. A number that divides evenly into another number is called a ________.

5. I can break a number into its prime factors step by step using a ________.

6 In 2^3, the small number 3 is the ____________ and it shows 2 is multiplied 3 times.

Reflect — Exit Ticket

What is the prime factorization of 40?

A. 2 × 2 × 2 × 5

B. 4 × 10

C. 5 × 8

D. 2 × 20

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor
5. **Notes 5:** Factor tree
6. **Notes 6:** Exponent
7. **Watch for this mistake:** *A common mistake in Prime Factorization is skipping the key idea: "Keep breaking a number apart until every factor is a prime number you cannot split anymore." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 36 — *$2^2 \times 3^2 = 4 \times 9 = 36$, so 36 is the code for this panel.*
9. **Try It 2:** A. $2 \times 3 \times 13$ — *$78 = 2 \times 39 = 2 \times 3 \times 13$. All three factors are prime, so $2 \times 3 \times 13$ unlocks the final panel.*
10. **Exit Ticket:** A. $2 \times 2 \times 2 \times 5$ — *$40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$. All factors (2, 2, 2, 5) are prime.*